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Eco 590 Proposal

# The Relationship Between Consumer Sentiment and Stock Market Volatility in the United States

## Introduction

Financial markets are influenced not only by economic fundamentals, but also by expectations and perceptions of future economic conditions. In recent decades, economists have increasingly emphasized the role of consumer and investor sentiment in shaping financial market outcomes. Measures of sentiment capture the degree of optimism or pessimism among consumers, which may influence investment decisions and market behavior.

This project proposes to examine the relationship between consumer sentiment and stock market volatility in the United States. By combining financial market data with macroeconomic indicators, this project aims to determine whether changes in sentiment are associated with fluctuations in market stability.

## Research Question

The primary research question is:

> **To what extent does consumer sentiment influence stock market volatility in the United States?**

This question will be explored using time-series data and regression analysis.

## Data Sources

This project will rely on publicly accessible Application Programming Interfaces (APIs) to collect data.

### Stock Market Data

Stock market data will be obtained from the Alpha Vantage API.

- Website: https://www.alphavantage.co
- Access: Open access with free API key
- Documentation: https://www.alphavantage.co/documentation/

The API provides daily financial data, including stock prices and trading volume. This study will focus on the SPDR S&P 500 ETF Trust (SPY), which serves as a proxy for overall U.S. stock market performance.

### Consumer Sentiment Data

Consumer sentiment data will be collected from the Federal Reserve Economic Data (FRED) API, by the Federal Reserve Bank of St. Louis.

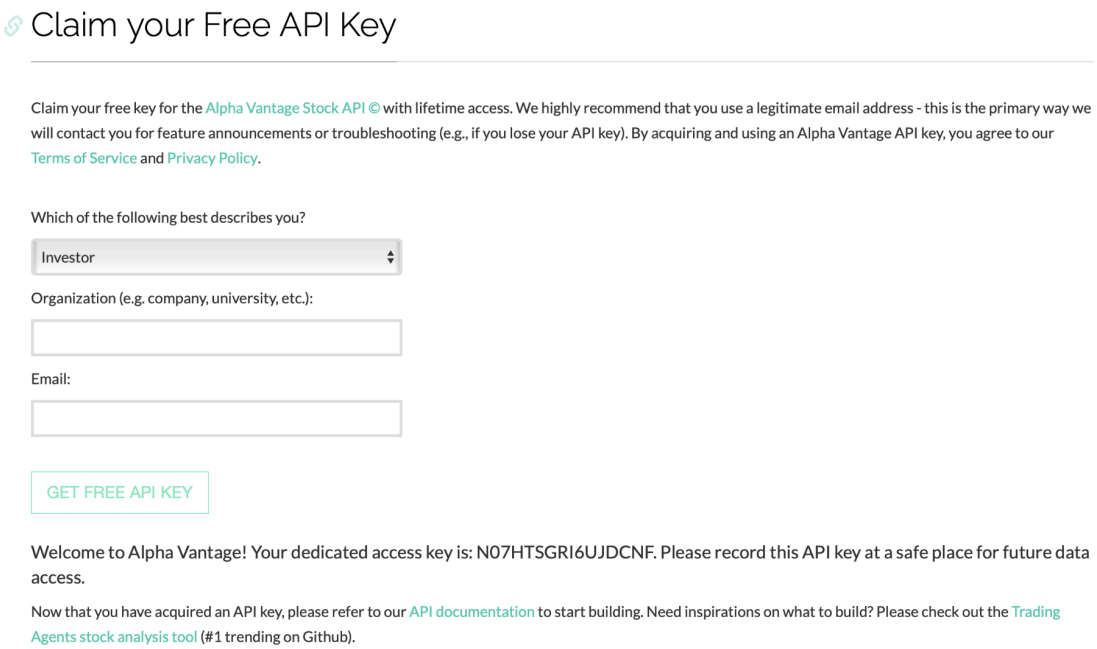
- Website: https://fred.stlouisfed.org/docs/api/api_key.html
- Access: Open access with free API key

This API provides macroeconomic indicators, including the University of Michigan Consumer Sentiment Index, which will be used to measure consumer confidence.

## API Access and Authorization

Both APIs used in this study are freely available for academic and non-commercial use.

The Alpha Vantage API requires users to register for a free API key. The documentation confirms that access is unrestricted for research purposes, subject to rate limits. FRED API also requires a free API key. The data is publicly available and widely used in academic research. Both APIs allow for academic usage,



**Variables**

This study will utilize three primary variables, including one constructed variable.

**Stock Market Returns**

Daily stock returns will be calculated using price data from the SPDR S&P 500 ETF Trust.

$$Return = \frac{P_t - P_{t-1}}{P_{t-1}}$$

**Consumer Sentiment Index**

Consumer sentiment will be measured using the University of Michigan Consumer Sentiment Index obtained from FRED.

This index reflects the degree of optimism or pessimism among consumers regarding current and future economic conditions.

**Stock Market Volatility (Constructed Variable)**

To satisfy the requirement of generating a non-standard variable, this study will construct a measure of stock market volatility.

Volatility will be calculated as the 30-day rolling standard deviation of daily stock returns. This measure captures the degree of variability in market returns over time and serves as an indicator of market uncertainty.

**Methodology**

Data will be collected and processed using Python, and the analysis will focus on data from 2010- 2025.

Steps that will be taken:

1. Retrieve daily stock price data from the Alpha Vantage API
2. Compute daily stock returns
3. Calculate 30-day rolling volatility
4. Obtain consumer sentiment data from FRED
5. Merge datasets by date
6. Clean and prepare the dataset for analysis

**Empirical Strategy**

The relationship between consumer sentiment and stock market volatility will be analyzed using both graphical and econometric methods.

The following visualizations will be created:

- Time-series plots of sentiment and volatility
- Scatter plots illustrating the relationship between the two variables

These graphs will provide an initial understanding of trends and correlations.

**Regression Model**

$$Volatility = \beta_0 + \beta_1(Sentiment) + \epsilon$$

Volatility represents the constructed volatility measure

Sentiment represents the consumer sentiment index

**Expected Results**

It is hypothesized that consumer sentiment will have a negative relationship with stock market volatility.

Specifically, lower sentiment (greater pessimism) is expected to be associated with higher volatility and higher sentiment (greater optimism) is expected to correspond to more stable markets.

**Conclusion**

This project aims to contribute to the understanding of how behavioral factors influence financial markets. By analyzing the relationship between consumer sentiment and stock market volatility, the study will provide insight into the role of expectations and psychological factors in shaping economic outcomes. The findings may be particularly relevant in periods of economic uncertainty, where shifts in sentiment can have significant effects on market behavior.